Redesigning Third-Year Biology Practicals for Research and Industry Preparedness

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Subject context

BIOL340: Cell and Molecular Biology

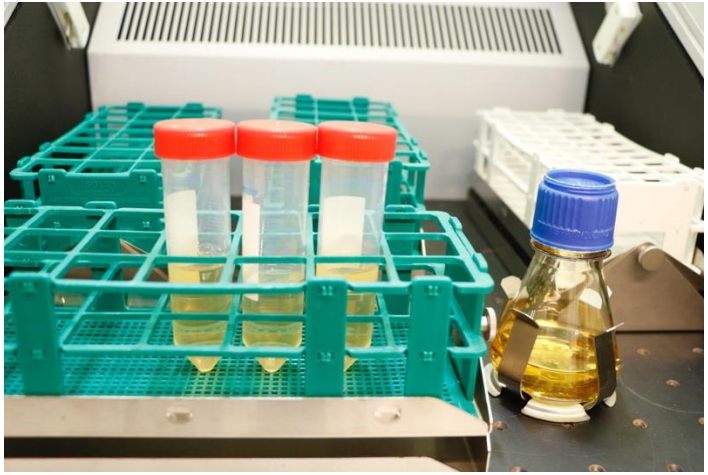
Subject coordinator: Dr Nikolas Johnston

- Examines the **molecular basis of cell structure and function** while providing hands-on training in key cell biology techniques, data analysis, and scientific communication

BIOl341: Immunity and Infection

Subject coordinator: Dr Emma-Jayne Proctor

- Explores the **interactions between microbial pathogens and the human immune system**, emphasizing infectious disease, diagnostics, therapeutics, and global health.



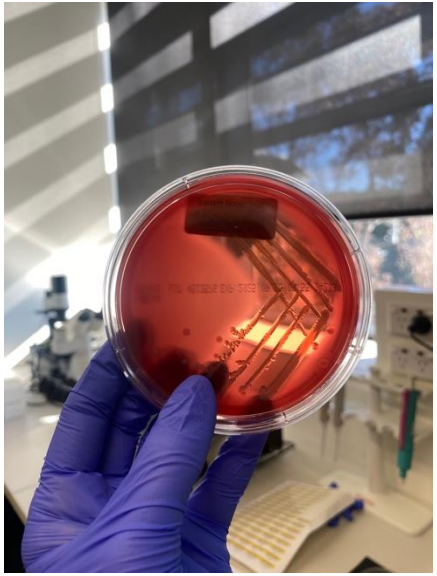
Other subjects:

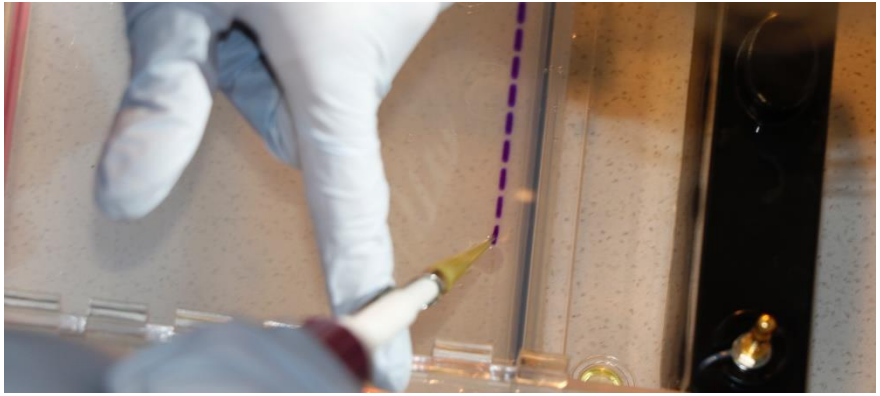
- BIOl342: Biomedical Research: Skills and Techniques
- BIOl343: Techniques in Biotechnology and Medical Research

Challenges

Rationale for Curriculum Redesign

1. BIOL340 and BIOL341 practicals **previously followed a week-by-week, skills-based format**.
   - This structure **limited opportunities** for students to link individual techniques to broader experimental objectives.
   - Students had **few opportunities to develop analytical and research-based skills** required for honours, higher-degree research, or industry work.
2. Some **complex quantitative assays were difficult for students to conceptualise** without complementary qualitative examples.
3. Maintaining **consistency in practical delivery across large student cohorts**





How it was addressed

1. Project-based practical format

Restructured from a week-to-week skills-based format to a project-based format, mimicking real-world research and industry scenarios.

Example: BIOL340 – Cell and Molecular Biology

Previous Format: Skills-Based Practicals

- Each practical focused on a **different standalone technique**.
- **No connection** between practicals or overarching scientific question.
- Focus on **technique acquisition** rather than experimental context or analysis.
- Included an **algae cell practical**, which did not align with the subject's focus on human and mammalian cell biology.

How it was addressed

1. Project-based practical format

Restructured from a week-to-week skills-based format to a project-based format, mimicking real-world research and industry scenarios.

Example: BIOL340 – Cell and Molecular Biology

Revised Format: Project-Based Practicals

- Practical classes have been **reframed around an authentic research scenario**, where students investigate and compare *healthy* and *diseased* cell populations. (Algae practical removed!)
- **Data and techniques built upon week by week**, allowing students to develop **progressively stronger evidence** of the differences between cells populations.
- Wet-labs were interspersed with **dry-labs sessions** for data analysis, discussion of results, and report writing, helping to synthesise meaning from their practical project.
- Work culminated in a **formal scientific report assessment**, reinforcing experimental design, analysis, and communication skills.

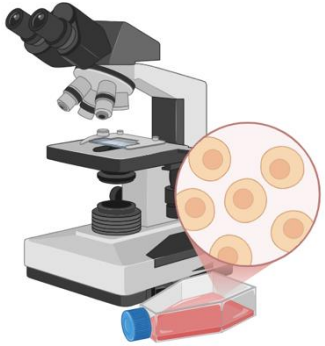
How it was addressed

1. Project-based practical format

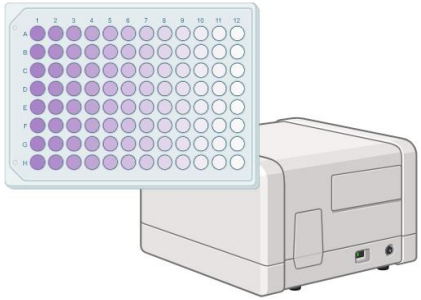
Restructured from a week-to-week skills-based format to a project-based format, mimicking real-world research and industry scenarios.

Example: BIOL340 – Cell and Molecular Biology

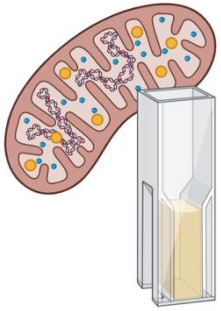
Revised Format: Project-Based Practicals



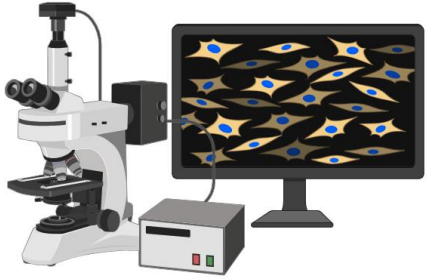
1. Basics of cell culture



2. Analysis of cell viability



3. Mitochondrial function

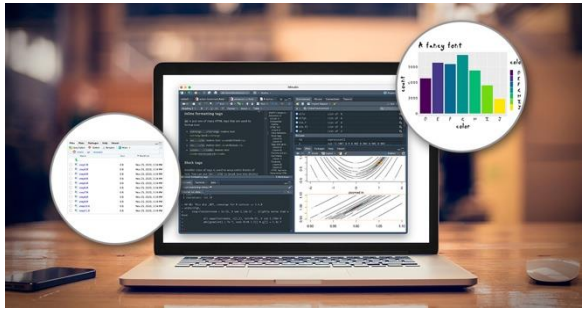


4. Apoptosis (cell death)

How it was addressed

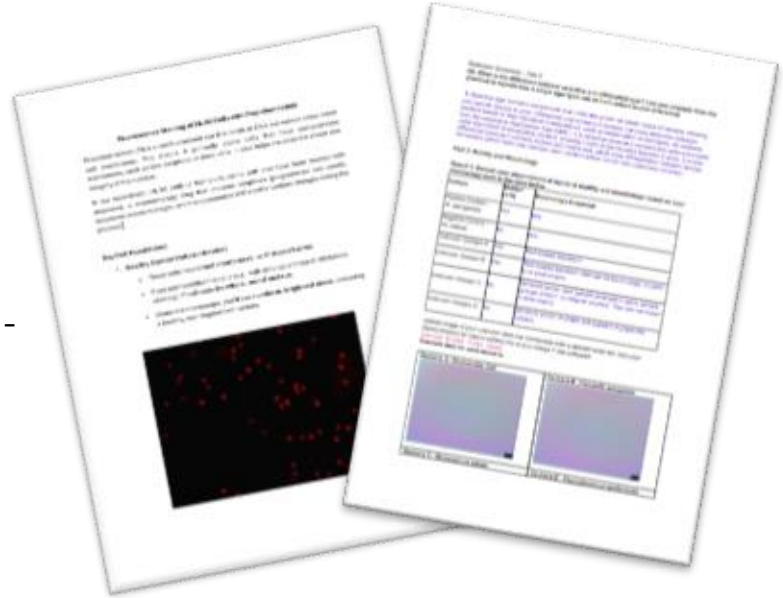
2. Digital enhancements

- BIOL340 included **shared online datasets** and the use of **R Studio** promoted collaborative and advanced data analysis.
- BIOL341 incorporated **online laboratory notebooks with interpretation and application questions to** strengthen comprehension and experimental reasoning



3. Enhanced demonstrator notes

- Development of **detailed demonstrator notes** to ensure consistency in teaching.
- Demonstrators **prepared for deviations** (e.g., strain-specific bacterial behaviour) **to support accurate explanations.**
- Approach enabled teaching through variability, encouraging class-wide comparison and critical interpretation.

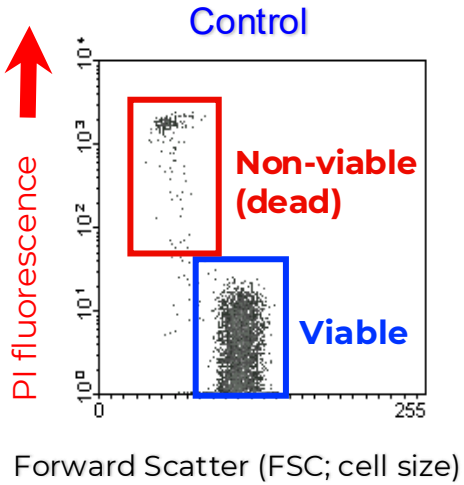


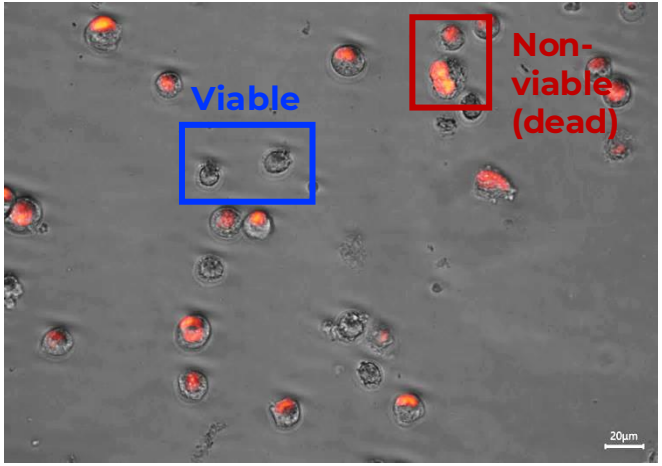
How it was addressed

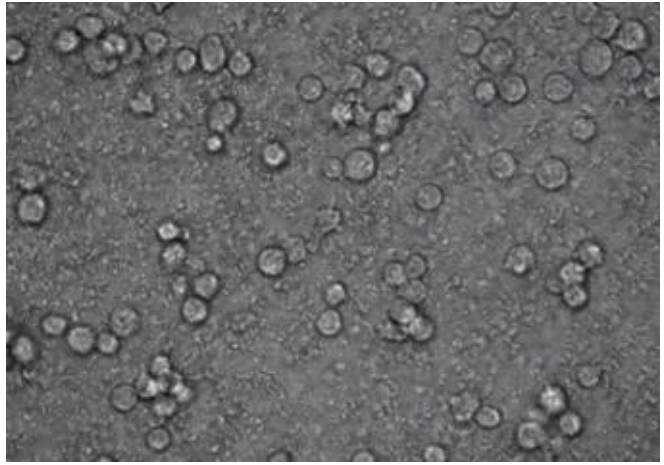
4. Improved resources to enhance understanding of complex techniques

Example: BIOL341 – Infection and Immunity

- Addition of user-friendly fluorescence microscope enabling **enhanced visualisation** to reinforce understanding
- Encourage best research practice by integrating **core experiment design principles**, such as having multiple methods of verification, and the use of appropriate controls.





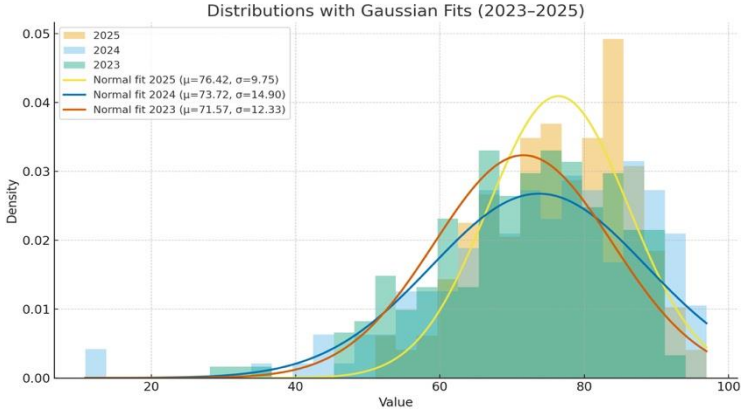


Learnings

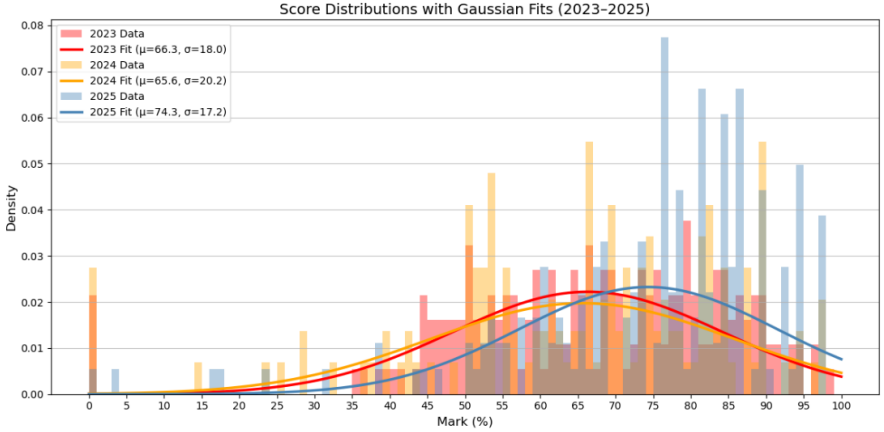
- **Student feedback**
  - Preliminary student feedback indicates greater engagement and appreciation of how experimental techniques interconnect within a research framework.
  - Students reported greater confidence in data interpretation, peer review, self-reflection, and scientific writing.

- **Improved student results**

BIOL340 – Cell and Molecular Biology



BIOL341 – Immunity and Infection



Next steps

1. Continue to improve practical class structure and in-class student/demonstrator resources.
2. Refine and expand the use of online notebooks and shared data systems to further enhance collaboration and reproducibility.
3. Extend project-based and research-aligned practical structures to additional molecular bioscience subjects.
4. Continue collaboration between academic and technical staff to identify further opportunities for improving teaching consistency and laboratory learning design.

Thank you for listening

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